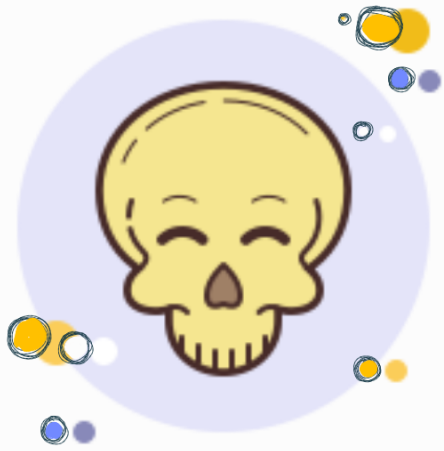


Set 10

Ecuații diferențiale de ordinul I

1.

(a) $x' = \frac{1}{t}x + t$, $t > 0$

I Rezolvăm ecuația omogenă atașată.

$$x' = \frac{1}{t}x \Rightarrow \frac{x'}{x} = \frac{1}{t} \Rightarrow \int \frac{x'}{x}\,dt = \int \frac{1}{t}\,dt$$

$$\Leftrightarrow \ln|x| = \ln|t| + C \Leftrightarrow |x| = e^C \cdot |t| \Leftrightarrow x = \pm e^C \cdot t$$

$$\Leftrightarrow x = k \cdot t$$

II Metoda variației constantelor

$x(t) = k(t) \cdot t \Rightarrow$

$$k' \cdot t + k = \frac{1}{t} \cdot k \cdot t + t \Rightarrow k' \cdot t + k = k + t \Rightarrow$$

$$k' \cdot t = t \Rightarrow k' = 1 \Rightarrow \int k'\,dt = t + C \Rightarrow k = t + C$$

$$\Rightarrow x(t) = (t + C)t = t^2 + Ct, \quad C \in \mathbb{R}$$

(b) $x' = e^t x + e^{2t}$

I Rezolvăm ecuația omogenă atașată

$$x' = e^t x \Leftrightarrow \frac{x'}{x} = e^t \Leftrightarrow \int \frac{x'}{x}\,dt = \int e^t\,dt$$

$$\Leftrightarrow \ln|x| = e^t + C \Leftrightarrow |x| = e^{e^t + C} \Leftrightarrow$$

$$|x| = e^C \cdot e^{e^t} \Leftrightarrow x = (\pm e^C) \cdot e^{e^t} \Leftrightarrow x = k \cdot e^{e^t},$$

$k \in \mathbb{R}$ constantă.

II Metoda variației constantelor

$x = k(t) \cdot e^{e^t} \Rightarrow$

$k' \cdot e^{e^t} + k \cdot e^{e^t} \cdot e^t = e^t k \cdot e^{e^t} + e^{2t} \Leftrightarrow$

$k' \cdot e^{e^t} = e^{2t} \Leftrightarrow k' = e^{2t - e^t} \Leftrightarrow$

$k = \int e^{2t - e^t} dt$

Notăm $e^t = u \rightarrow \frac{du}{dt} = e^t \rightarrow dt = e^{-t} du,$

$t = \ln|u| \Rightarrow$

$k = \int u \cdot e^{-u} du = -e^{-u} \cdot u + \int e^{-u} du = -e^{-u} \cdot u - e^{-u} + C$

$= -e^{-e^t} \cdot e^t - e^{-e^t} + C \Rightarrow$

$x = (-e^{-e^t} \cdot e^t - e^{-e^t} + C) \cdot e^{e^t} = \boxed{-e^t - 1 + C \cdot e^{e^t}}$

(C) $(1+t^2)x' - 2tx = t$

I Rezolvăm ecuația omogenă atașată

$(1+t^2)x' - 2tx = 0 \Leftrightarrow (1+t^2)x' = 2tx \Leftrightarrow$

$x' = \frac{2t}{1+t^2} \cdot x \Leftrightarrow \frac{x'}{x} = \frac{2t}{1+t^2} \Leftrightarrow \int \frac{x'}{x} dt = \int \frac{2t}{1+t^2} dt$

$\Leftrightarrow \ln|x| = \ln(1+t^2) + C \Leftrightarrow |x| = (1+t^2) \cdot e^C \Leftrightarrow$

$x = (\pm e^C) \cdot (1+t^2) \Leftrightarrow x = k \cdot (1+t^2)$, unde k este o constantă.

II Metoda variației constantelor

$x = k(t) \cdot (1+t^2)$

$\Rightarrow (1+t^2)(k'(1+t^2) + k \cdot 2t) - 2t(k \cdot (1+t^2)) = t \Leftrightarrow k'(1+t^2) +$

$2kt + k'(1+t^2)t^2 + k \cdot 2t^3 - 2tk - 2kt^3 = t \Leftrightarrow$

$k'(t^4 + 2t^2 + 1) = t \Leftrightarrow k'(t^2+1)^2 = t \Leftrightarrow k' = \frac{t}{(t^2+1)^2}$

$\Leftrightarrow k = \int \frac{t}{(t^2+1)^2} dt \Leftrightarrow k = -\frac{1}{2} \cdot \frac{1}{(t^2+1)} + C \rightarrow$ constantă

$x = (-\frac{1}{2} \cdot \frac{1}{t^2+1} + C)(1+t^2) = -\frac{1}{2} + C(1+t^2)$, $C \in \mathbb{R}$

(e) $x' \cos t - x \sin t = 1$

I Rezolvăm ecuația omogenă atașată

$x' \cos t - x \sin t = 0 \Leftrightarrow x' \cos t = x \sin t \Leftrightarrow x' = x \operatorname{tg} t \Leftrightarrow$

$\frac{x'}{x} = \operatorname{tg} t \Leftrightarrow \int \frac{x'}{x} dt = \int \operatorname{tg} t \, dt \Leftrightarrow \ln|x| = \int \frac{\sin t}{\cos t} dt \Leftrightarrow$

$\ln|x| = -\ln|\cos t| + C \Leftrightarrow \ln|x| = \ln \frac{1}{|\cos t|} \cdot e^C \Leftrightarrow$

$|x| = \frac{1}{|\cos t|} \cdot e^C \Leftrightarrow x = \pm e^C \cdot \frac{1}{\cos t} \Leftrightarrow x = K \cdot \frac{1}{\cos t}$

II Metoda variației constantelor

$x(t) = K(t) \cdot \frac{1}{\cos t} \Rightarrow$

$\left(K' \cdot \frac{1}{\cos t} + \frac{\sin t}{\cos t} \cdot K\right) \cos t - K \cdot \frac{1}{\cos t} \sin t = 1 \Leftrightarrow$

$K' + \frac{\sin t}{\cos t} K - K \cdot \frac{\sin t}{\cos t} = 1 \Leftrightarrow K' = 1 \Leftrightarrow K = t + C \Rightarrow$

$x(t) = (t+C) \cdot \frac{1}{\cos t} = \frac{t}{\cos t} + \frac{C}{\cos t} = \frac{C+t}{\cos t}$, $C \in \mathbb{R}$ constantă

d) $x' - \frac{2x}{t} = t^2 \cdot e^t$, $t > 0$

I Rezolvăm ecuația omogenă atașată

$x' - \frac{2x}{t} = 0 \Leftrightarrow x' = \frac{2x}{t} \Leftrightarrow \frac{x'}{x} = \frac{2}{t} \Leftrightarrow$

$\int \frac{x'}{x} dt = \int \frac{2}{t} dt \Leftrightarrow \ln|x| = 2 \ln|t| + C \Leftrightarrow$

$|x| = t^2 \cdot e^C \Leftrightarrow x = \pm e^C t^2 \Leftrightarrow x = K \cdot t^2$

II Metoda variației constantelor

$x(t) = K(t) \cdot t^2 \Rightarrow$

$K' \cdot t^2 + K \cdot 2t - \frac{2t^2 K}{t} = t^2 \cdot e^t \Leftrightarrow K' t^2 = t^2 \cdot e^t \Leftrightarrow$

$K' = e^t \Leftrightarrow K = e^t + C \Rightarrow$

$x(t) = (e^t + C) t^2 = \boxed{e^t \cdot t^2 + t^2 C}$, $C \in \mathbb{R}$ constantă

f) $(t-t^3)x' + (2t^2-1)x - t^3 = 0$

I Rezolvăm ecuația omogenă atașată

$(t-t^3)x' + (2t^2-1)x = 0 \Leftrightarrow (t-t^3)x' = (1-2t^2)x \Leftrightarrow$

$\Leftrightarrow \frac{x'}{x} = \frac{1-2t^2}{t-t^3} \Leftrightarrow \frac{x'}{x} = \frac{1-2t^2}{t(1-t^2)} \Leftrightarrow$

$\int \frac{x'}{x}dt = \int \frac{1-2t^2}{t(1-t^2)}dt \Leftrightarrow \ln|x| = \int \left(\frac{1-t^2}{t(1-t^2)} - \frac{t^2}{t(1-t^2)}\right)dt$

$+C$

$\Leftrightarrow \ln|x| = \ln|t| - \int \frac{t}{1-t^2}dt \Leftrightarrow \ln|x| = \ln|t| + \frac{1}{2}\ln|1-t^2|$

$\Leftrightarrow \ln|x| = \ln|t\cdot\sqrt{1-t^2}\cdot e^C| \Leftrightarrow |x| = |t|\cdot\sqrt{|1-t^2|}\cdot e^C$

$\Leftrightarrow x = \pm e^C \cdot t\sqrt{|1-t^2|} \Leftrightarrow x = k\cdot t\sqrt{|1-t^2|}$ → constantă

II Metoda variației constantelor

$x = k(t)\cdot t\cdot\sqrt{|1-t^2|}$

$(t-t^3)\left(k'\cdot t\sqrt{1-t^2} + k\sqrt{|1-t^2|} + k\cdot t\cdot\frac{-t}{\sqrt{|1-t^2|}}\right)$

$+(2t^2-1)kt\sqrt{|1-t^2|} = t^3 \Leftrightarrow (t-t^3)k't\sqrt{|1-t^2|}$

$+\cancel{(t-t^3)k\sqrt{|1-t^2|}} + (t-t^3)\cancel{kt}\frac{-t}{\sqrt{|1-t^2|}} + (2t^2-1)\cdot$

$\cdot kt\cancel{\sqrt{|1-t^2|}} = t^3 \Leftrightarrow (t-t^3)k'\cancel{\sqrt{|1-t^2|}} = t^2 \Leftrightarrow$

$\cancel{(1-t^2)}k'\sqrt{|1-t^2|} = t^{\cancel{2}} \Leftrightarrow k' = \frac{t}{(1-t^2)\sqrt{|1-t^2|}} \Leftrightarrow$

$k' = \frac{t}{|1-t^2|^{\frac{3}{2}}} \Leftrightarrow k' = \frac{1}{\sqrt{|1-t^2|}} + C$

$\Rightarrow x = \left(\frac{1}{\cancel{\sqrt{|1-t^2|}}} + C\right)t\cdot\sqrt{|1-t^2|} = \boxed{t + t\sqrt{|1-t^2|}\cdot C}$

g) $x' + 2tx = 2te^{-t^2}$

I Rezolvăm ecuația omogenă atașată

$x' + 2tx = 0 \Leftrightarrow x' = -2tx \Leftrightarrow \frac{x'}{x} = -2t \Leftrightarrow$

$\int \frac{x'}{x}\,dt = \int -2t\,dt \Leftrightarrow \ln|x| = -t^2 + C \Leftrightarrow |x| = e^{-t^2}\cdot e^C$

$\Leftrightarrow x = (\pm e^C)\cdot e^{-t^2} \Leftrightarrow x = k\cdot e^{-t^2}$, $k\in\mathbb{R}$ constantă

II Metoda variației constantelor

$x(t) = k(t)\cdot e^{-t^2}$

$k'\cdot e^{-t^2} + k\cdot -e^{-t^2}\cdot 2t + 2tk e^{-t^2} = 2t\cdot e^{-t^2}$

$\Leftrightarrow k' = 2t \Leftrightarrow k = t^2 + C \Rightarrow$

$X(t) = (t^2+C)\cdot e^{-t^2}$

(h) $x'\sin t - x = 1 - \cos t$, $t\in(0,\pi)$

I Rezolvăm ecuația omogenă atașată

$x'\sin t - x = 0 \Leftrightarrow x'\sin t = x \Leftrightarrow \frac{x'}{x} = \frac{1}{\sin t} \Leftrightarrow$

$\Leftrightarrow \int \frac{x'}{x}\,dt = \int \frac{1}{\sin t}\,dt \Leftrightarrow \ln|x| = \int \frac{1-\cos t+\cos t}{\sin t}\,dt \Leftrightarrow$

$\ln|x| = \int \frac{2\sin^2\frac{t}{2}}{\sin t}\,dt + \int \frac{\cos t}{\sin t}\,dt \Leftrightarrow \int \frac{2\sin^2\frac{t}{2}}{2\sin\frac{t}{2}\cos\frac{t}{2}}\,dt +$

$\int \frac{(\sin t)'}{\sin t}\,dt = \ln|x| \Leftrightarrow \ln|x| = \int \frac{\sin\frac{t}{2}}{\cos\frac{t}{2}}\,dt + \ln|\sin t| + C$

$\Leftrightarrow -2\ln\left|\cos\frac{t}{2}\right| + \ln|\sin t| + C = \ln|x| \Leftrightarrow |x| = \frac{|\sin t|}{\cos^2\frac{t}{2}}\cdot e^C$

$\Leftrightarrow x = k\cdot\frac{\sin t}{\cos^2\frac{t}{2}}$

II Metoda variației constantelor

$x(t) = k(t)\cdot\frac{\sin t}{\cos^2\frac{t}{2}} \Rightarrow$

$x'\sin t - x = 1-\cos t \Leftrightarrow \left(k'\cdot\frac{\sin t}{\cos^2\frac{t}{2}} + k\cdot\left(\frac{\sin(\frac{t}{2})\sin t}{\cos^3(\frac{t}{2})} + \frac{\cos t}{\cos^4\frac{t}{2}}\right)\right)$

$\cdot \sin t - k\cdot\frac{\sin t}{\cos^2\frac{t}{2}} = 1-\cos t \Leftrightarrow$

$\frac{\sin(\frac{t}{2})\cdot 2\sin(\frac{t}{2})\cos(\frac{t}{2})}{\cos^3(\frac{t}{2})} + \frac{\cos^2\frac{t}{2} - \sin^2\frac{t}{2}}{\cos^2\frac{t}{2}} =$

$= \frac{2\sin^2\frac{t}{2} + \cos^2\frac{t}{2} - \sin^2\frac{t}{2}}{\cos^2\frac{t}{2}} = \frac{\sin^2\frac{t}{2} + \cos^2\frac{t}{2}}{\cos^2\frac{t}{2}} = \frac{1}{\cos^2\frac{t}{2}} \Rightarrow$

$x' - \sin t - x = 1 - \cos t \Leftrightarrow k' \frac{\sin^2 t}{\cos^2 \frac{t}{2}} = 1 - \cos t \Leftrightarrow$

$k' = \frac{(1-\cos t)\cos^2 \frac{t}{2}}{\sin^2 t} \Leftrightarrow k' = \frac{2\sin^2 \frac{t}{2} \cdot \cos^2 \frac{t}{2}}{4 \sin^2 \frac{t}{2} \cos^2 \frac{t}{2}} \Leftrightarrow k' = \frac{1}{2} \Leftrightarrow k = t \cdot \frac{1}{2} + C$

$\Rightarrow x(t) = (\frac{t}{2} + C) \cdot \frac{\sin t}{\cos^2 \frac{t}{2}} = (\frac{t}{2} + C) \cdot \frac{2 \sin \frac{t}{2} \cos \frac{t}{2}}{\cos^2 \frac{t}{2}} =$

$(C + \frac{t}{2}) \cdot \operatorname{tg} \frac{t}{2} \cdot 2 = (\frac{C}{2} + t) \cdot \operatorname{tg} \frac{t}{2} = (C_1 + t) \cdot \operatorname{tg} \frac{t}{2}$, $C_1 \in \mathbb{R}$ constantă.

(i) $tx' + (t+1)x = 3t^2 \cdot e^{-t}$, $t > 0$

I Rezolvăm ecuația omogenă atașată.

$tx' + (t+1)x = 0 \Leftrightarrow tx' = -(t+1)x \Leftrightarrow \frac{x'}{x} = \frac{-(t+1)}{t}$

$\Leftrightarrow \ln|x| = -\int \frac{t+1}{t} dt \Leftrightarrow \ln|x| = -t - \ln|t| + C \Leftrightarrow$

$|x| = e^C \cdot \frac{1}{|t|} \cdot e^{-t} \Leftrightarrow x = k \cdot \frac{1}{t \cdot e^t}$

II Metoda variației constantelor

$x(t) = k(t) \cdot \frac{1}{t \cdot e^t} \Rightarrow$

$tx' + (t+1)x = 3t^2 e^{-t} \Leftrightarrow t(k' \cdot \frac{1}{te^t} + k \cdot \frac{-(t+1)e^{-t}}{t^2}) +$

$(t+1) \frac{1}{te^t} \cdot k = 3t^2 \cdot e^{-t} \Leftrightarrow k' \cdot \frac{t}{te^t} = 3t^2 \cdot e^{-t} \Leftrightarrow$

$k' = 3t^2 \Leftrightarrow k = t^3 + C \Rightarrow$

$x(t) = (t^3 + C) \cdot \frac{1}{te^t} = \frac{C + t^3}{t} \cdot e^{-t}$, $C \in \mathbb{R}$ constantă

(j) $x' + x \operatorname{ctg} t = e^t$, $t \in (0, \pi)$

I Rezolvăm ecuația omogenă atașată

$x' + x \operatorname{ctg} t = 0 \Leftrightarrow x' = x \cdot -\operatorname{ctg} t \Leftrightarrow \frac{x'}{x} = \frac{-\cos t}{\sin t} \Leftrightarrow$

$\ln|x| = -\ln|\sin t| + C \Leftrightarrow x = k \cdot \frac{1}{\sin t}$

II Metoda variației constantelor

$x(t) = k(t) \cdot \frac{1}{\sin t} \Rightarrow$

$x' + x \operatorname{ctg} t = e^t \Leftrightarrow k' \cdot \frac{1}{\sin t} + k \cdot \frac{-\cos t}{\sin^2 t} + k \cdot \frac{1}{\sin t} \cdot \frac{\cos t}{\sin t} = e^t$

$\Leftrightarrow k' \cdot \frac{1}{\sin t} = e^t \Leftrightarrow k' = \sin t \, e^t$

$k = \int \sin t \, e^t dt = e^t \sin t - \int \cos t \, e^t dt = e^t \sin t - \cos t \, e^t - \int e^t \sin t \, dt \Rightarrow k = \frac{1}{2}(e^t \sin t - e^t \cos t) + C$

$\Rightarrow x(t) = \left[\frac{1}{2} e^t (\sin t - \cos t) + C\right] \frac{1}{\sin t} = \frac{1}{2} e^t (1 - \operatorname{ctg} t) + \frac{C}{\sin t}$,

$C \in \mathbb{R}$ constantă.

(K) $x' + 3t^2 x = 6t^2$

I Rezolvăm ecuația omogenă atașată

$x' + 3t^2 x = 0 \Leftrightarrow x' = -3t^2 x \Leftrightarrow \frac{x'}{x} = -3t^2 \Leftrightarrow$

$\ln|x| = \int -3t^2 dt \Leftrightarrow \ln|x| = -t^3 + C \Leftrightarrow |x| = K \cdot e^{-t^3}$

II Metoda variației constantelor

$x(t) = K(t) \cdot e^{-t^3} \Rightarrow k' e^{-t^3} + k \cdot (-3t^2) e^{-t^3} + 3t^2 \cdot k \cdot e^{-t^3} = 6t^2$

$\Leftrightarrow k' = 6t^2 \cdot e^{t^3} \Leftrightarrow k = 2e^{t^3} + C \Rightarrow$

$x = (2e^{t^3} + C) \cdot e^{-t^3} = 2 + C \cdot e^{-t^3}$, $C \in \mathbb{R}$ constantă

(L) $x' = t \sin 2t + x \operatorname{tg} t$, $t \in (-\frac{\pi}{2}, \frac{\pi}{2})$

I Rezolvăm ecuația omogenă atașată

$x' = x \operatorname{tg} t \Leftrightarrow \frac{x'}{x} = \operatorname{tg} t \Leftrightarrow \ln|x| = \int \operatorname{tg} t \, dt \Leftrightarrow$

$\ln|x| = -\ln|\cos x| + C \Leftrightarrow x = k \cdot \frac{1}{\cos t}$

II Metoda variației constantelor

$x(t) = k(t) \frac{1}{\cos t} \Rightarrow$

$x' = t \sin 2t + x \operatorname{tg} t \Leftrightarrow k' \frac{1}{\cos t} + k \frac{\sin t}{\cos^2 t} = t \sin 2t + k \frac{\sin t}{\cos^2 t}$

$\Leftrightarrow k' = t \sin 2t \cos t \Leftrightarrow k = \frac{2((\cos^2 t + 2)\sin t - 3t \cos^3 t)}{9} + C$

$\Rightarrow x(t) = \frac{1}{\cos t}\left(C + \frac{2((\cos^2 t + 2)\sin t - 3t\cos^3 t)}{9}\right) =$

$= \frac{C}{\cos t} + \frac{1}{9}(2 \operatorname{tg} t (\cos^2 t + 2) - 3t \cos^2 t)$

(2) Nu avem ecuații Bernoulli la examen!

(3) Metoda aproximărilor succesive

(b) $x' = 2t(x+1)\,,\ x(0) = 0$

$x_0(t) = 0$

$x_1(t) = 0 + \int_0^t 2s(0+1)\,ds = 0 + s^2\Big|_0^t = t^2$

$x_2(t) = 0 + \int_0^t 2s(s^2+1)\,ds = t^2 + \frac{s^4}{2}\Big|_0^t = t^2 + \frac{t^4}{2}$

Să demonstrăm prin inducție că

$P(k): x_{n+1}(t) = \sum_{k=1}^{n} \frac{t^{2k}}{k!}$

I $x_1(t) = t^2$ (A)

II Presupunem P(k) adevărat până la un k oarecare fixat $\Rightarrow$

$x_k(t) = \sum_{i=1}^{k} \frac{t^{2i}}{i!}$

$\Rightarrow x_{k+1}(t) = 0 + \int_0^t 2s\left(\sum_{i=1}^{k} \frac{s^{2i}}{i!} + 1\right) ds = \left(s^2 + \sum_{i=1}^{k} \frac{2s^{2i+2}}{(i+1)!\,2}\right)\Big|_0^t =$

$= t^2 + \sum_{i=1}^{k} \frac{t^{2(i+1)}}{(i+1)!} = \sum_{i=1}^{k+1} \frac{t^{2i}}{i!} \overset{I+II}{\Rightarrow}$ conform principiul I

al inducției matematice $x_n = \sum_{i=1}^{n} \frac{t^i}{i!} \Rightarrow$

$x = \lim_{n\to+\infty} x_n = \lim_{n\to+\infty} \sum_{i=1}^{n} \frac{x^i}{i!} = \sum_{n=1}^{\infty} \frac{t^{2n}}{n!} = e^{t^2} - 1$